Big Data and Machine Learning for Applied Economics

Week 08: The Papers and the Synthesis of the Case

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Previously... and today

- Seven sessions, one case (the 2022 polling stations), and one workflow, from the split to reporting where it fails.
- Last week, representation: what varies most is not what predicts. The census's PC1 captures 32.0 % of its variance and explains 0.9 % of the vote.
- The question that never changed: not what the effect of X on Y is, but **how well we can predict Y and what that depends on**. Today it gets answered.

8:05–9:35	Six papers, six 15-minute presentations
9:35–9:45	Break
9:45–10:10	The synthesis of the case
10:10–10:50	Project pitches: the one-page idea
10:50–11:00	Optional closing: the causal bridge, with our data

No new method today

Today we use all of them. Each paper answers the same seven questions; the synthesis is seeing them together.

1 The presentations

2 The synthesis of the case

3 The project pitches

4 Optional closing: the causal bridge

Six areas of economics, six papers

Each team picked its area and paper on September 22 [choices and presenters: to be filled in after session 2]

Area	**Option 1**	**Option 2**
Political economy	Kim & Zilinsky (2024, *Political Behavior*)	Bazzi et al. (2022, *REStat*)
Crime and justice	Kleinberg et al. (2018, *QJE*)	Angelova, Dobbie & Yang (2026, *REStud*)
Health	Mullainathan & Obermeyer (2022, *QJE*)	Einav et al. (2018, *Science*)
Development and poverty	Aiken et al. (2023, *Nature*)	Haushofer et al. (2025)
Labor and education	Hoffman, Kahn & Li (2018, *QJE*)	Chalfin et al. (2016, *AER P&P*)
Finance and credit	Fuster et al. (2022, *J. Finance*)	Dobbie et al. (2021, *REStat*)
Macro and forecasting	Medeiros et al. (2021, *JBES*)	Goulet Coulombe et al. (2022)
IO and demand	Dubé & Misra (2023, *JPE*)	Bandiera et al. (2020, *JPE*)

Six of these areas have had owners since session 2; the full menu, with summaries, is in the `literature/` index. Today the papers enter the course as protagonists for the first time: until now, the case was the only thread.

The 15-minute script, and how it is graded (10% of the course)

The script (twelve minutes of presentation, three of questions):

1. The question, and the decision that depends on the prediction.
2. The data and the unit of observation; what is known at prediction time.
3. The benchmark it is measured against.
4. How it is evaluated out of sample, and where the model fails.
5. The main result, in its units.
6. Which method from the course it is, and what we would have done differently.

The reading is graded, not the slide design:

- **Precision:** what is predicted, with what data, against which benchmark, and on which test set; one headline number with its units.
- **Protocol:** split and validation; whether they respect time, space, and groups; where leakage could have crept in.
- **Interpretation and critique:** what the model says and what it does not; whether any conclusion treats an importance as an effect; what we would have done differently.

Twelve minutes means twelve minutes, and questions are answered with the paper in hand.

While you listen: each paper's scorecard

One row per paper: it feeds the synthesis and models what your project must answer.

Field	**The question**
y and unit	What is predicted, and for whom (person, station, country–year, village)?
Decision	What changes if the prediction is good? Who uses it?
Benchmark	What is it compared against: the mean, a simple rule, the judge, night lights?
Validation	Split, CV, holdout, recursive forecast? Does it respect time, space, and groups?
Method	Which family from the course, and why that one?
Open box	Which session 6 tool do they use, and what do they claim with it?
Where it fails	For whom, when, and outside which range does it degrade?

If a field stays empty, that is your question for the presenter. Each presentation ends with a question from the room on point 4 or point 6 of the script: prepare it while you listen.

1 The presentations

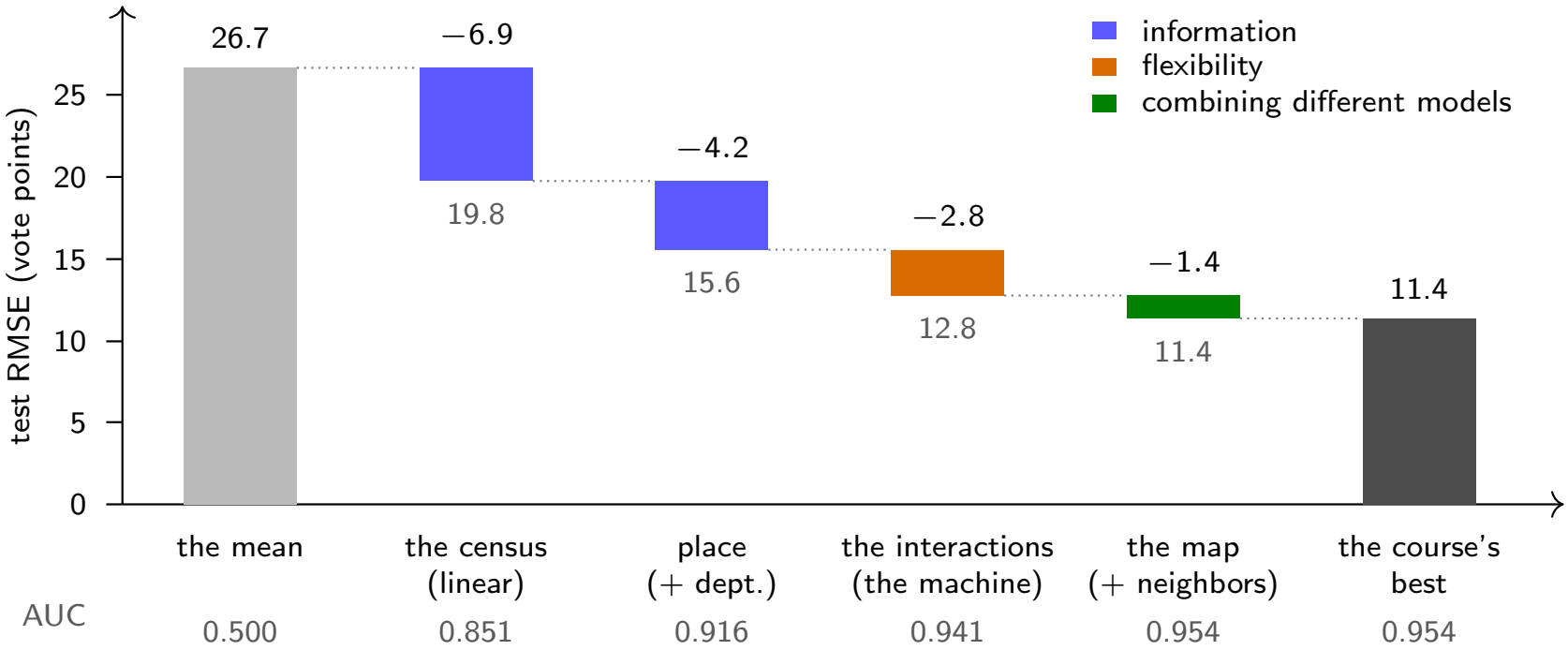
2 The synthesis of the case

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The course's question, answered: from 26.7 to 11.4

Each step adds one thing, in the order the course added it; the same 2,398 test stations



- Of the 15.3 points gained over the mean, 11.1 (73 %) came from **information** —the census and place—, 2.8 from **flexibility** —the interactions the machine found on its own— and 1.4 from **combining two models** that err differently. Classification climbs the same staircase (the AUC, below).
- The split depends on the order, like Shapley's (session 6): with the census alone, boosting already beats the line by 4.1 points (15.7 versus 19.8).

The course's table: information and flexibility

Test RMSE for Petro's vote share: moving down a row adds information; moving right, flexibility

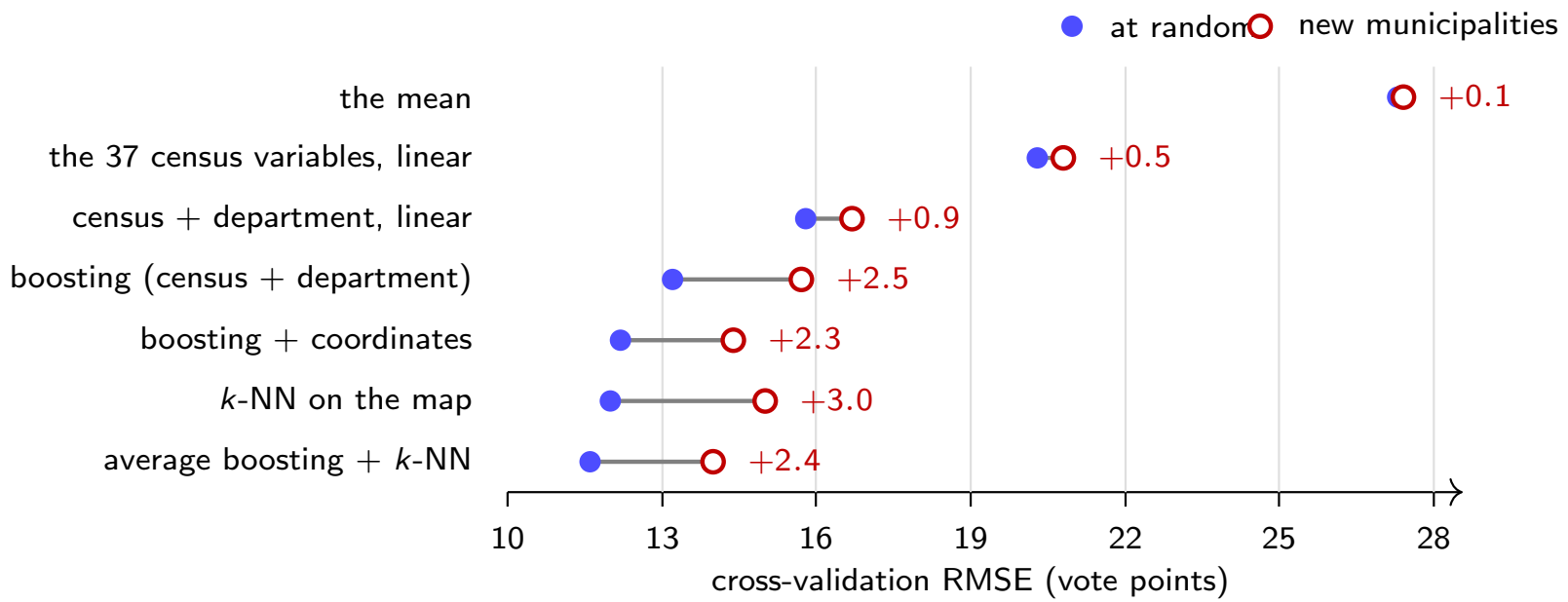
Ingredients	Linear	Tree	Forest	Boosting	Network	Neighbors
The 37 census variables	19.8	19.1	15.6	15.7	16.9	
+ department	15.6	15.7	12.9	12.8	13.6	
The map (coordinates)		13.7				12.0
Everything together		14.4	12.1	12.1	13.2	

The benchmark, the mean: 26.7. The course's best, the average of boosting (no map) and the neighbors: **11.4** (adding the network, 11.5).

- **Moving down a row is worth as much as moving right:** the department in a straight line (15.6) leaves the same as a forest without it (15.6).
- **With the map, the families converge** (forest and boosting 12.1, neighbors 12.0) and the network trails (13.2): on tabular data, trees rule (session 7).
- **In does-Petro-win, the same staircase:** the linear census AUC is 0.851; with the department, 0.916; forest and boosting, 0.943 and 0.941; with the map, 0.950; the average, 0.954.

Does the structure travel? To new municipalities

The same ten-fold judge: stations assigned at random, or whole municipalities held out (sessions 3 and 5)



- **Everyone loses, and most of all whoever leaned hardest on the neighbors:** k-NN on the map (+3.0) drops from second to third place and boosting with coordinates beats it. The average stays first (14.0).
- Each scheme answers its own question: random folds, "how do I do at stations like the ones I saw?"; whole municipalities, "how do I do in a new municipality?".

Does the structure travel? To other groups and another year

- **To other places, no:** Cali's champion line (session 2) did not travel to the rest of Valle or to the country, where education alone explains 0.04 % of the vote.
- **To other groups, halfway** (the boosting opened in session 6): the rural error (14.9) is 1.6 times the urban one (9.3); by department it runs from 7.4 in Atlántico to 26.9 in Guaviare (7 stations); and the extremes get compressed: where Petro got under 20, the model predicts 20.1 (actual, 12.5).
- **To another year, halfway:** the 2018 vote *ranks* almost perfectly (AUC 0.95) but *decides* with an old cutoff (84.3 % accuracy). And which election is "more predictable" depends on the metric: accuracy says 2010 (0.901 versus 0.832); the AUC, 2022 (0.906 versus 0.878).

The sentence that sums up the case

Predictability is a property *of the territory and the moment*, not of the algorithm. In your projects: say *where* you want the model to travel —other places, other groups, another year— and validate *that*.

The workflow, with names attached

Each step of the pipeline, with the week of the case that burned it in:

Step	**Where we lived it**
1. Split	W2–W3: the fixed split (9.602 and 2.398); the test set, once per week
2. Preprocess inside the fold	W3: the municipal vote computed on everyone promised 11.2 and was worth 13.0; W7: PCA within each fold
3. Validate according to the structure	W3 and W5: at random versus whole municipalities
4. Tune on the training set	W3–W7: λ, k, pruning, m, rounds, units, by CV or OOB
5. Evaluate against the benchmark	W2–W4: the mean and "wins where he won in 2018"
6. Interpret with caveats	W6: permutation, ablation, PDP, and SHAP question the model
7. Report where it fails	W6: rural or urban, the extremes, one department or another

For the project: these seven steps are the sections of the November 24 document. Today's papers are proof that the same workflow carries a publishable article.

The course in one slide

1. **Y, not β.** Predicting and explaining are different questions; the training error is optimistic.
2. **The line as a prediction machine:** the benchmark, the canonical table, predicted versus observed, the error by subgroup; and the line does not travel.
3. **The judge and the knob:** cross-validation decides how much complexity; Ridge divides, Lasso subtracts, and the Lasso selects predictors, not causes.
4. **Classifying is estimating a probability and deciding:** the model ranks, calibration says whether to believe it, and costs set the cutoff.
5. **Split, average, and correct:** trees find the interactions on their own; averaging and adding in sequence fixes their variance.
6. **Opening the box:** importance, partial dependence, and Shapley question the model, not the world.
7. **To represent is to compress with purpose:** what varies most is not necessarily what predicts.

The golden rules, one last time

#1 The training error is optimistic. #2 The test set is used only once. #3 CV validates entire pipelines. #4 The benchmark is what the decision would do without the model ("wins where he won in 2018").

1 The presentations

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The one-page idea and the five-minute pitch

Project deliverable 1: 10 of the 100 points

The page (sent before class) answers three things:

1. **The question and the decision** behind it: who would use the prediction and what they would do differently with it.
2. **The unit and the y:** what exactly is predicted, for which unit, and at what moment.
3. **The candidate data:** source, approximate size, availability, and what would be known *at prediction time*.

The pitch: five minutes, at most three slides, followed by two minutes of feedback from the room.

The three questions the room always asks

What decision changes with the prediction? What information exists at prediction time, and which is leakage in disguise? Against what benchmark and with what test will you know it worked?

Una propuesta que no sobrevive a esas tres preguntas hoy es mejor que una que las descubre el 24 de noviembre.

From the idea to the November 24 document

Deliverable 2: a document of at most 8 pages (50 points) and a 20-minute presentation (40 points)

The document's sections, which are the workflow's steps:

1. The question and the decision.
2. The data: source, unit, size, predictors by blocks, what would be known at prediction time.
3. Metric, benchmark, and a validation strategy that matches the data's structure (groups, time, space).
4. The methods to compare and why: at least two families from the course.
5. What would be interpreted and what would not.
6. Risks: leakage, extrapolation, regime change, fairness if people are involved.
7. Work plan.

- It is a **proposal**, not a report: no results are required, only a design that withstands the seven questions. A preliminary table adds, but does not replace, the design.
- [Individual or in pairs: TBD.] The November 24 presentation lasts 20 minutes plus 10 of questions.

1. The presentations
2. The synthesis of the case
3. The project pitches
4. **Optional closing: the causal bridge**

The causal bridge (I): when a prediction machine helps estimate a β

The partially linear model: $y = \theta\, d + g(x) + \varepsilon$, where θ is the effect of interest, d the treatment, and x many controls of unknown shape.

- **What does not work:** a Lasso of y on (d, x), then reading d's coefficient. The penalty biases $\hat{\theta}$ and drops controls that matter little for y but a lot for d: the Lasso selects predictors, not causes (session 3).
- **Double selection** (Belloni, Chernozhukov & Hansen, 2014): select the controls that predict y *and* those that predict d, take the union, and run OLS with all of them. The machine picks controls; the inference is OLS.

The division of labor

ML predicts the *nuisances* (the part of y and of d the controls explain); the economist supplies the identification assumption (that x is enough). Without it, no algorithm turns a predictor into a cause.

The causal bridge (II): residuals on residuals

Double machine learning (Chernozhukov et al., 2018)

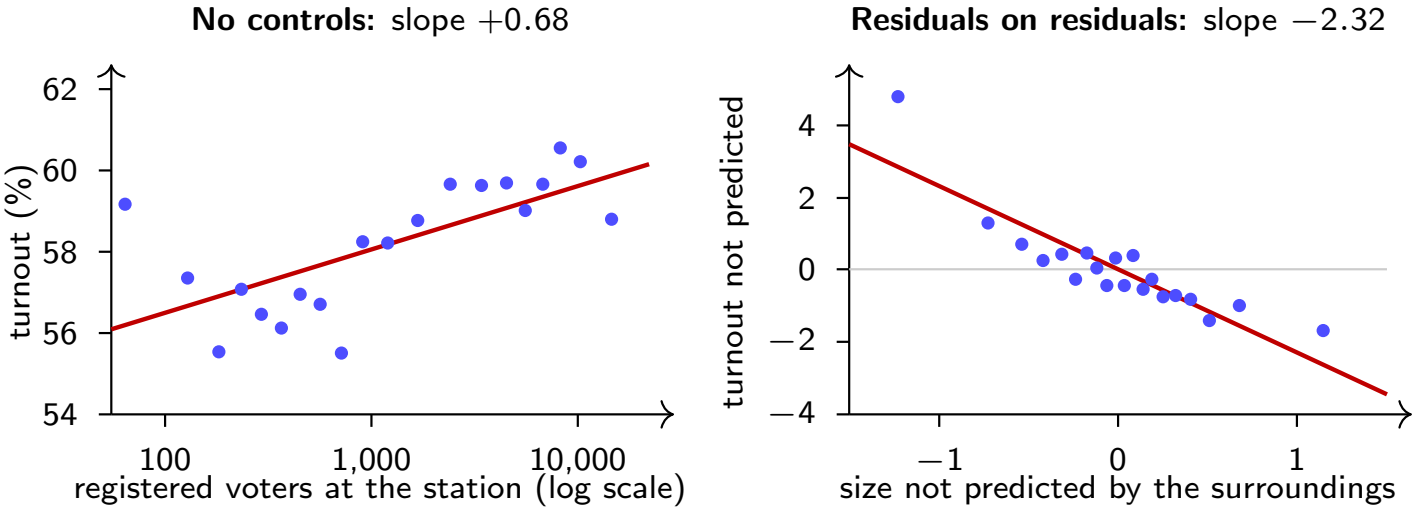
Predict y from x and d from x with *any* method from the course, in crossed folds, and regress one residual on the other:

$$\hat{\theta} = \frac{\sum_i \big(d_i - \hat{m}(x_i)\big)\big(y_i - \hat{\ell}(x_i)\big)}{\sum_i \big(d_i - \hat{m}(x_i)\big)^2}, \qquad \hat{m}(x) \approx \mathbb{E}[d \mid x], \;\; \hat{\ell}(x) \approx \mathbb{E}[y \mid x].$$

- **It is Frisch–Waugh–Lovell with different machines:** with linear $\hat{m}$ and $\hat{\ell}$, $\hat{\theta}$ is OLS with controls. With boosting, the controls enter in whatever shape the data demand.
- **Cross-fitting:** each residual comes from a model that *never saw* that station (session 3's CV, in the service of a β); and the estimator is insensitive, to first order, to small errors in the two predictions (appendix A1).
- **With our data:** do larger stations have lower turnout? d = log of registered voters; y = turnout; x = the composition of the surroundings (32 census columns), the department, and the coordinates.

The causal bridge (III): do large stations vote less?

11,984 stations; standard errors clustered by municipality; out-of-fold R^2

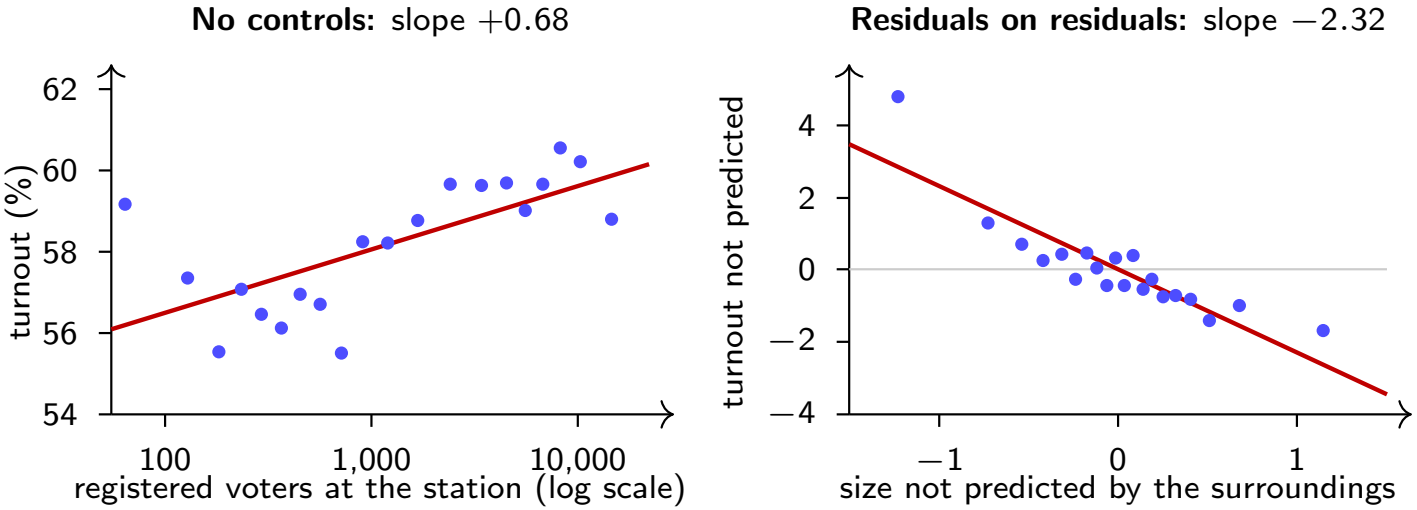


	$\hat{\theta}$	R^2: turnout	R^2: size
No controls	+0.68 (0.35)		
OLS, controls in a straight line	−1.67 (0.29)	0.44	0.72
DML with boosting	−2.32 (0.28)	0.55	0.89

- Doubling a station's size, holding the surroundings fixed: about 1.6 points less turnout. Boosting predicts size better (0.89 versus 0.72) and the effect moves away from the line's.
- **Question for the room:** what story would give the same sign without size causing anything?

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- **Question for the room:** what story would give the same sign without size causing anything? Registrants who no longer live there inflate the rolls and lower turnout by construction. DML does not fix that: it is identification, not prediction (A3).

What is left of the semester

Tuesday, November 10	No class. Problem Set 2 due at 11:59 p. m. Optional office hours during class time for the problem set and the project.
Tuesday, November 17	No class.
Tuesday, November 24	Project deliverable 2: the document, that same day. 20-minute presentations plus questions. Fifteen opening minutes with the Problem Set 2 ranking and the course closing.

For the document: today's seven steps, the scorecards you filled out this morning, and the three papers closest to your question as reporting models. The project's horse races (Gu, Kelly & Xiu, 2020; Medeiros et al., 2021) are the reading for anyone comparing many families.

For Problem Set 2: with session 7 you have everything you need. The course's final model is chosen by validation, refit on the full training set, and tested once.

Essential references

- The papers from the areas menu are in `literature/`, organized by folder, with the index (`indice.csv`): authors, journal, methods, and why they are in the course.
- Belloni, A., Chernozhukov, V., & Hansen, C. (2014). High-Dimensional Methods and Inference on Structural and Treatment Effects. *JEP*, 28(2), 29–50.
- Chernozhukov, V., et al. (2018). Double/Debiased Machine Learning for Treatment and Structural Parameters. *Econometrics Journal*, 21(1), C1–C68.
- Wager, S., & Athey, S. (2018). Estimation and Inference of Heterogeneous Treatment Effects Using Random Forests. *JASA*, 113(523), 1228–1242.
- Athey, S., & Imbens, G. (2019). Machine Learning Methods That Economists Should Know About. *Annual Review of Economics*, 11, 685–725.
- Gu, S., Kelly, B., & Xiu, D. (2020). Empirical Asset Pricing via Machine Learning. *RFS*, 33(5), 2223–2273. [A horse-race template for the project.]
- Medeiros, M., Vasconcelos, G., Veiga, Á., & Zilberman, E. (2021). Forecasting Inflation in a Data-Rich Environment. *JBES*, 39(1), 98–119.
- Gelvez, J. D., Cardiles, A., Martínez-González, E. F., & Muñoz, M. (2026). How Predictable Is an Election? Working paper. [The source of the case data.]

5 Appendix: going deeper

A1. Why residuals-on-residuals works

Frisch–Waugh–Lovell, without the formula: d's coefficient is the same if you first strip from y and from d what x explains of each, then regress one residual on the other. With linear g that is OLS; with unknown g you must *predict* $\mathbb{E}[y \mid x]$ and $\mathbb{E}[d \mid x]$, and that is where the course comes in.

Two ingredients from Chernozhukov et al. (2018):

1. **Neyman orthogonality.** The estimator is insensitive, to first order, to small errors in $\hat{m}$ and $\hat{\ell}$: the two prediction errors multiply each other, and the product of two small things is smaller. The machines only need to converge faster than $n^{-1/4}$, not to be perfect.
2. **Cross-fitting.** If $\hat{m}$ and $\hat{\ell}$ are fit on the same observations that later estimate θ, the machine's overfit leaks into $\hat{\theta}$. Split the sample into K folds and predict each fold's nuisances with models fit on the others: golden rule #3.

- **Causal forests** (Wager & Athey, 2018): "honest" trees (one subsample picks the splits, another estimates the effect in each leaf) for heterogeneous effects $\theta(x)$. With an identification assumption, always.

A2. The bridge, inside: the sample, the controls, and the nuisances

- **The sample:** out of the 12,000 stations, 16 with more votes than eligible voters are dropped (none with more than 97 registered): they are recording errors, not turnout.
- **The controls** (67 columns): the composition of the surroundings (education, age, ethnicity, services, stratum, households, distance, rurality), the department, and the coordinates. Out go the size columns —population, dwellings, and blocks, with correlation 0.91 with d— and the gap between voting-age population and registered voters, which is built from the treatment: controlling for them would be controlling for d.
- **The nuisances:** session 5's boosting ($d = 6$, $\eta = 0.05$), with rounds picked by a five-fold judge within each training set (between 252 and 604); 60 fits in total. Five different partitions give -2.35 to -2.22; the median is reported, -2.32 (s.e. 0.28 clustered by municipality; 0.22 unclustered). With linear nuisances, the same DML gives -1.70: the line with controls.

A3. The bridge, inside: what moves the number

- **The 16 dropped stations:** with them, OLS gives -2.14 and DML -3.46. 16 out of 12,000 stations move the DML by 1.1 points: look at the data before estimating.
- **The stations much smaller than their surroundings predict** (the 5 % with the lowest $d - \hat{m}(x)$: 600 stations, median 105 registered, 80 % rural, turnout 62.1 versus 58.2): without them, the DML gives -1.59. They are the first point in the figure.
- **The sign holds; the magnitude, not entirely:** it depends on a few small stations. That fragility belongs to the data, not the method.
- **The rival story** (the question for the room): the gap would measure registrants who no longer live there, but it is built from the treatment. No available control rules it out: that is why identification is an assumption.